Exp 8 : Page replacement algorithm : FIFO

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lab-03-17@lab-03-17-OptiPlex-3280-AIO:~$ gedit cp3_1290.c
^C
lab-03-17@lab-03-17-OptiPlex-3280-AIO:~$ cat cp3_1290.c
#include <stdio.h>
#include <unistd.h>

int main() {
    int forkid1, forkid2, forkid3;

    forkid1 = fork();

    if (forkid1 < 0) {
        printf("First fork failed\n");
    }
    else if (forkid1 == 0) {
        // Child process 1
        printf("Child process 1 PID %d, parent PID %d\n", getpid(), getppid());
    }
    else {
        forkid2 = fork();

        if (forkid2 < 0) {
            printf("Second fork failed\n");
        }
        else if (forkid2 == 0) {
            // Child process 2
            printf("Child process 2 PID %d, parent PID %d\n", getpid(), getppid());
        }
        else {
            forkid3 = fork();

            if (forkid3 < 0) {
                printf("Third fork failed\n");
            }
            else if (forkid3 == 0) {
                // Child process 3
                printf("Child process 3 PID %d, parent PID %d\n", getpid(), getppid());
            }
            else {
                // Parent process
                printf("Parent process PID %d\n", getpid());
            }
        }
    }

    return 0;
}
lab-03-17@lab-03-17-OptiPlex-3280-AIO:~$ gcc cp3_1290.c
lab-03-17@lab-03-17-OptiPlex-3280-AIO:~$ ./a.out
Child process 1 PID 6788, parent PID 6787
Parent process PID 6787
Child process 2 PID 6789, parent PID 6787
Child process 3 PID 6790, parent PID 6787
```

```
lab-03-17@lab-03-17-OptiPlex-3280-AIO: ~
#include <stdio.h>

int main() {
    int frames, pages;

    printf("Enter number of frames: ");
    scanf("%d", &frames);

    printf("Enter number of pages: ");
    scanf("%d", &pages);

    int pageRef[pages], memory[frames];
    int i, j, k, flag;
    int pageFaults = 0, pageHits = 0;
    int index = 0;

    printf("Enter page reference string:\n");
    for(i = 0; i < pages; i++) {
        scanf("%d", &pageRef[i]);
    }

    // Initialize frames
    for(i = 0; i < frames; i++) {
        memory[i] = -1;
    }

    // Header
    printf("\n%-8s", "Page");
    for(i = 0; i < frames; i++) {
        printf("F%-4d", i + 1);
    }
    printf("%-8s\n", "Status");

    // FIFO Logic
    for(i = 0; i < pages; i++) {
        flag = 0;

        for(j = 0; j < frames; j++) {
            if(memory[j] == pageRef[i]) {
                flag = 1;
                pageHits++;
                break;
            }
        }

        if(flag == 0) {
```

lab-03-17@lab-03-17-OptiPlex-3280-AIO: ~

```c
    // Header
    printf("\n%-8s", "Page");
    for(i = 0; i < frames; i++) {
        printf("F%-4d", i + 1);
    }
    printf("%-8s\n", "Status");

    // FIFO Logic
    for(i = 0; i < pages; i++) {
        flag = 0;

        for(j = 0; j < frames; j++) {
            if(memory[j] == pageRef[i]) {
                flag = 1;
                pageHits++;
                break;
            }
        }

        if(flag == 0) {
            memory[index] = pageRef[i];
            index = (index + 1) % frames;
            pageFaults++;
        }

        printf("%-8d", pageRef[i]);

        for(k = 0; k < frames; k++) {
            if(memory[k] != -1)
                printf("%-5d", memory[k]);
            else
                printf("%-5s", "-");
        }

        if(flag == 0)
            printf("%-8s\n", "Fault");
        else
            printf("%-8s\n", "Hit");
    }

    printf("\nPage Faults = %d\n", pageFaults);
    printf("Page Hits   = %d\n", pageHits);

    return 0;
}
```